

Cybersecurity: Endpoints Devices Vulnerability, Attacks, and Defenses

Finding vulnerabilities on devices that are connected to a network, how these devices can be exploited and how to defend/secure against those attacks is the core of cybersecurity. Your goal in this module is to learn those skills, but first you need to be comfortable with the terms associated with those topics.

Please provide a **short and concise** definition/explanation for each of the cybersecurity terminologies. Each definition/explanation entry is worth **5** points.

Direction & Submission: use the downloaded document, provide the definition/explanation for each indicated row below. At the completion, submit the completed document through Canvas. You have up to **two weeks** to submit the assignment after the due date. Please note that there is a daily 3% deduction if submitted after the due date.

	Terms	Definition/explanation
1	Workstations	are personal computing devices used for professional or technical tasks, typically featuring higher performance, reliability, and security than standard consumer computers
2	Clients	are devices or software applications that request services, resources, or data from a server within a networked environment
3	endpoints	are devices such as computers, mobile phones, servers, or IoT devices that connect to a network and serve as access points for communication and data exchange
4	malware	is malicious software designed to disrupt, damage, or gain unauthorized access to computer systems, networks, or data.
5	Ransomware	is a type of malware that encrypts a victim's data and demands payment, usually in cryptocurrency, to restore access
6	Blocking ransomware	involves implementing security measures such as endpoint protection, regular backups, software updates, network segmentation, and user awareness training to prevent ransomware attacks and mitigate their impact.
7	Locking ransomware	is a type of malware that restricts access to a device or system by locking the screen and demanding payment to restore functionality.
	Eavesdrop (to secretly listen to a target)	
8	Keylogger	is a type of malware or hardware device that records keystrokes on a computer or mobile device to steal sensitive information such as passwords and financial data.
9	Spyware	is malicious software that secretly monitors and collects user activity, including keystrokes, browsing habits, and personal information, often for malicious purposes.
	Masquerade (an attempt to deceive or hide identity)	
10	Trojan	is a type of malware that disguises itself as legitimate software to trick users into installing it, allowing attackers to gain unauthorized access or cause harm to a system.

11	Remote Access Trojan (RAT)	type of malware that allows attackers to remotely control an infected device, often without the user's knowledge, enabling them to steal data, install additional malware, or monitor activity.
	Launching attacks to other devices	
12	Virus	type of malware that attaches itself to a legitimate program or file and spreads to other programs or files when executed
13	File-based virus	type of virus that attaches itself to executable files or documents and spreads when the infected file is opened or executed
14	Fileless Virus	type of malware that operates in memory and does not rely on files or software to infect a system, making it harder to detect using traditional antivirus tools
15	Living-off-the-land binaries (LOLBins)	are legitimate system tools or software that are exploited by attackers to carry out malicious activities
16	Worm	self-replicating type of malware that spreads across networks without the need for human interaction
17	Bloatware	refers to software that is pre-installed on a device, often consuming excessive system resources with unnecessary features or programs that slow down the device and are typically unwanted by the user.
18	Bot	is an automated software program designed to perform repetitive tasks, often maliciously used in botnets to launch attacks like DDoS or spread spam.
19	Logic bomb	type of malware that is triggered by a specific event or condition
20	Rootkits	type of malware designed to hide the existence of certain processes, files, or system data from detection, allowing attackers to maintain privileged access to a compromised system.
21	backdoor	method of bypassing normal authentication or security measures to gain unauthorized access to a system
	Application Vulnerabilities and attacks (<i>vulnerabilities in installed applications</i>)	
22	Privilege escalation	is the process of exploiting a vulnerability or misconfiguration in a system to gain higher-level access or privileges
23	Buffer overflow attack (<i>injection/memory injections</i>)	occurs when a program writes more data to a buffer than it can hold, causing excess data to overwrite adjacent memory
24	Improper Exception and Error Handling	occurs when a program fails to properly manage runtime errors

25	Null pointer	is a pointer in programming that does not point to any valid memory location or object
26	Race condition	occurs when multiple processes or threads access shared resources simultaneously in an unsynchronized manner,
27	Time of check (TOC) to time of use (TOU)	refers to a type of vulnerability where the state of a resource is checked for validity (TOC) but changes before it is used (TOU), allowing an attacker to exploit the time gap and alter the resource, potentially leading to unauthorized actions.
	Application Attacks	
28	Web-based attacks	malicious activities that target web applications or websites to exploit vulnerabilities
29	Directory traversal	is a type of attack where an attacker manipulates a web application's input to gain access to files and directories that are outside the intended directory, potentially exposing sensitive data or system files.
30	Scripting Attack	is a type of web-based attack where malicious scripts (often JavaScript) are injected into a website or web application, which then execute on a user's browser to steal data, execute actions, or manipulate content.
31	Cross-site scripting (XSS)	is a vulnerability where attackers inject malicious scripts into web pages viewed by other users
32	Injection Attack	is a type of attack where malicious data is inserted into an application, causing it to execute unintended commands or queries
33	SQL injection (SQLi)	is an attack where malicious SQL code is inserted into a query, allowing attackers to manipulate the database, retrieve unauthorized data, or execute harmful commands.
34	Request Forgery Attack	occurs when an attacker tricks a user into making a request that performs an unwanted action on a web application where the user is authenticated
35	Cross-site Request Forgery (CSRF)	is an attack where a malicious website or application tricks a user into performing an unintended action on a trusted site where they are authenticated
36	Server-site Request Forgery (SSRF)	is an attack where an attacker manipulates a server to make requests to internal or external resources that the server can access
37	Replay	is a type of attack where valid data transmission is intercepted and retransmitted by an attacker to gain unauthorized access or perform malicious actions

	Protecting endpoints	
38	Antivirus (AV)	is a software designed to detect, prevent, and remove malicious software (malware) such as viruses, worms, and trojans, helping to protect computers and networks from potential threats.
39	Static analysis	is the process of examining a program's code, without executing it, to identify potential vulnerabilities, bugs, or security issues, typically performed during the development phase to improve software quality and security.
40	Dynamic analysis	is the process of analyzing a program while it is running, in order to identify real-time behavior, performance issues, and security vulnerabilities
41	Web Browsers	are software applications used to access, retrieve, and view content on the internet, allowing users to browse websites, interact with web applications, and display multimedia content.
42	Secure cookies	are cookies that are transmitted over secure connections (HTTPS) and are marked with the "Secure" flag, ensuring that they are only sent over encrypted channels to protect them from interception or tampering during transmission.
43	HTTP Response Headers	are metadata sent by a server in response to an HTTP request, providing information about the server, the response, and how the browser should handle the content
44	Monitoring and response to systems (<i>network endpoints</i>)	refers to the continuous observation and analysis of network devices (like computers, servers, or IoT devices) to detect potential security threats, performance issues, or anomalies, and taking appropriate actions
45	Host Intrusion Detection Systems (HIDS)	are security systems installed on individual devices (hosts) that monitor and analyze activity for suspicious behavior, potential intrusions, or unauthorized access, providing alerts and logs to help protect the system from security threats.
46	Host Intrusion Prevention Systems (HIPS)	are security solutions installed on individual devices that not only detect potential intrusions but also take proactive measures to prevent or block malicious activity
47	Endpoints detection and response (EDR)	refers to a security approach that focuses on monitoring, detecting, and responding to suspicious activities and threats on endpoint devices
48	Hardening the endpoints	involves implementing security measures to strengthen endpoint devices
49	Patch Management	is the process of identifying, acquiring, testing, and installing software updates (patches) to fix vulnerabilities, improve functionality, and ensure the security of systems

		and applications, minimizing the risk of exploitation by attackers.
50	patching	refers to the process of applying updates or fixes to software, systems, or applications to correct vulnerabilities, enhance security, and improve performance, thereby reducing the risk of security breaches and system failures.
51	patch distribution (<i>Automated patch management tools</i>)	refers to the use of software tools that automatically distribute and apply patches across multiple systems and endpoints
52	patch reception	refers to the process in which a system or device receives patches or updates from a centralized source or vendor
53	Operating Systems	are software platforms that manage hardware resources and provide services for computer programs, enabling users to interact with the system and run applications.
54	Disabling unnecessary ports and protocols	is a security practice that involves turning off unused or non-essential communication ports and protocols on systems
55	Application Allow List	is a security practice where only pre-approved, trusted applications are allowed to run on a system
56	Sandbox	is a security mechanism that isolates programs or processes in a controlled environment, allowing them to run without affecting the host system, preventing potential harm from untrusted or malicious code.